



COAL MINE SAFETY OVERVIEW (1997-2022)

An Analysis of Indian Coal Mines Accidents

Year

All



15705

Total Accidents



2053

Total Fatal Accid...



2430

Total Deaths



13652

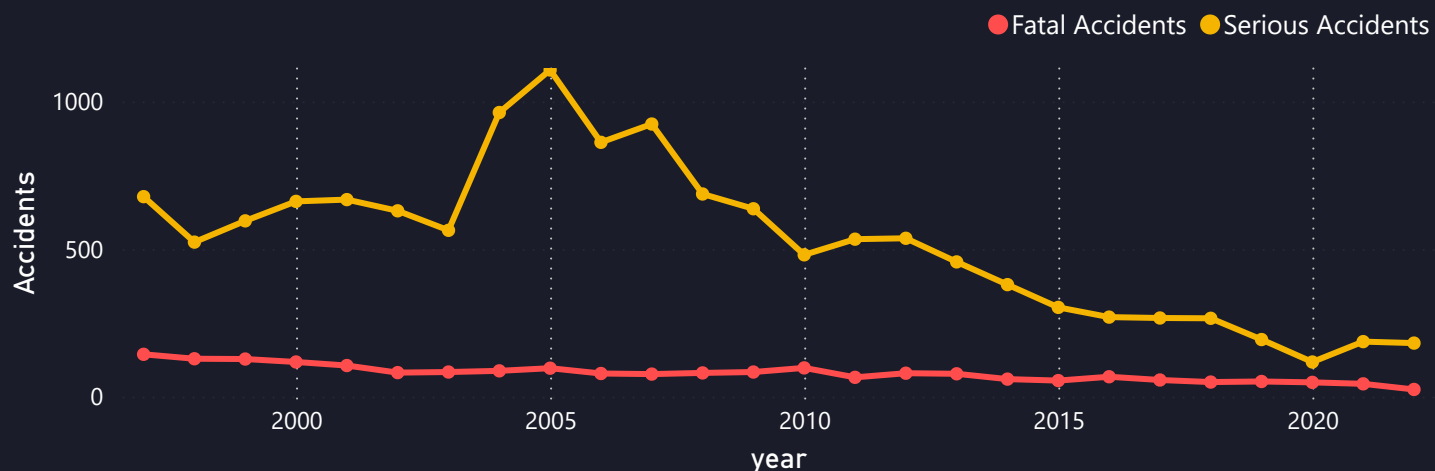
Total Serious Acc...



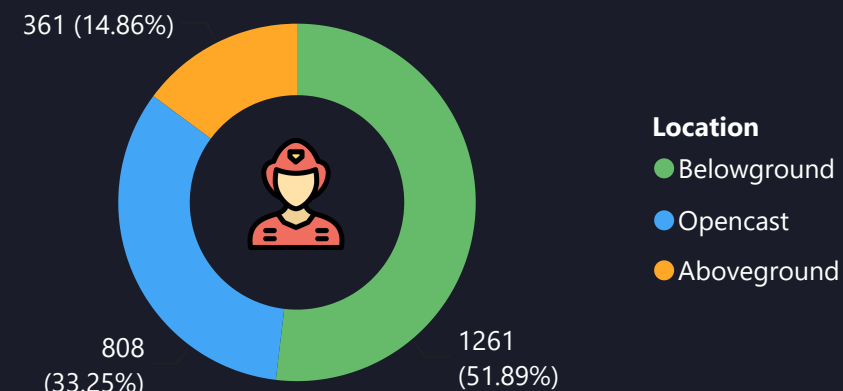
16735

Total Person Affe...

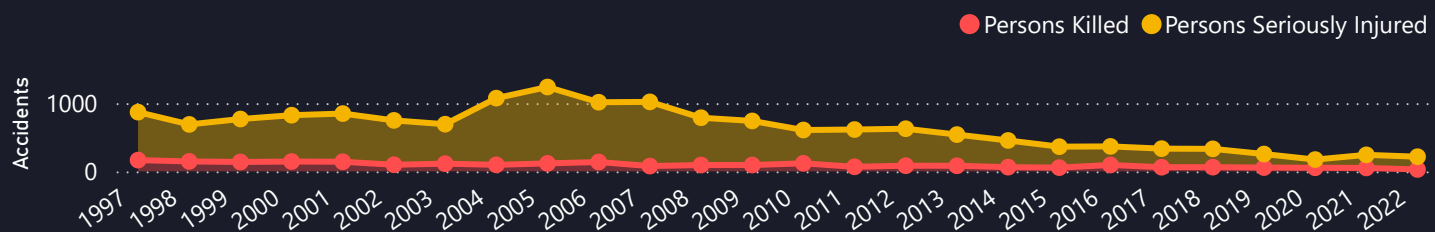
Fatal + Serious Accident Trend (1997–2022)



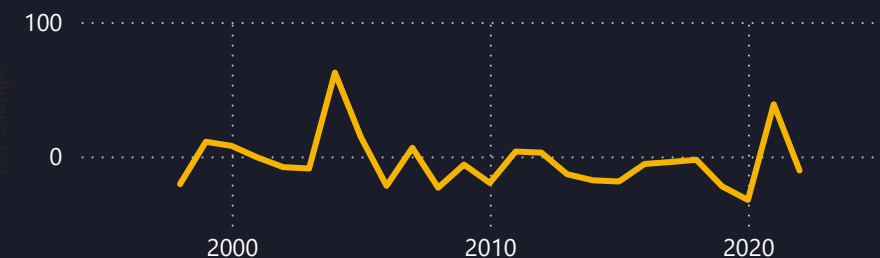
Fatality Share by Location



Total Persons Affected: Killed vs Seriously Injured



Year-on-Year % Change in Total Accidents





SAFETY RISK & ENFORCEMENT ANALYSIS

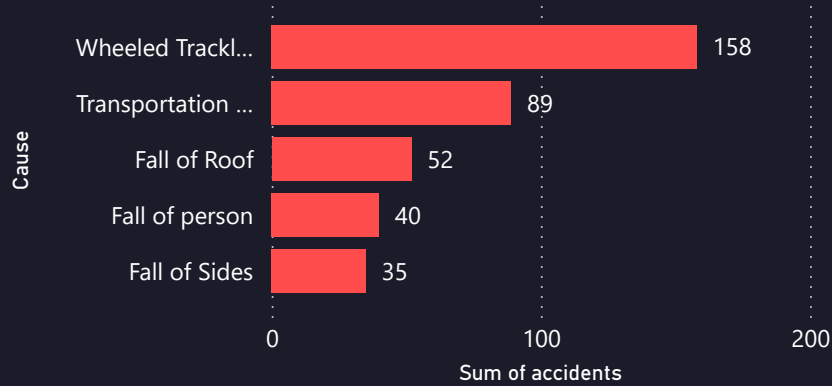
Exploring Accident Causes, Risk Patterns, and Regulatory Interventions in Indian Coal Mines

Cause

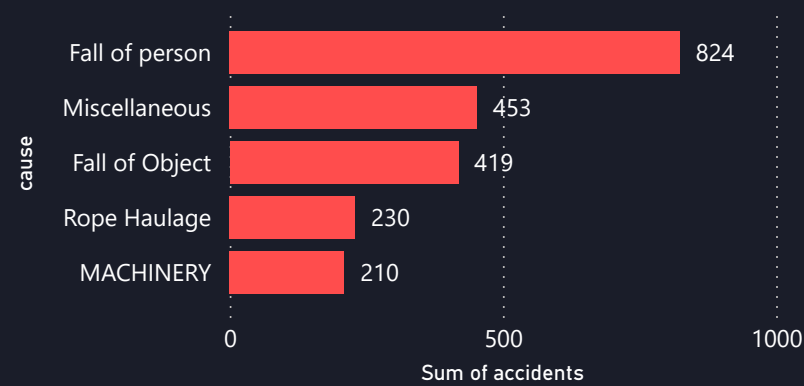
All



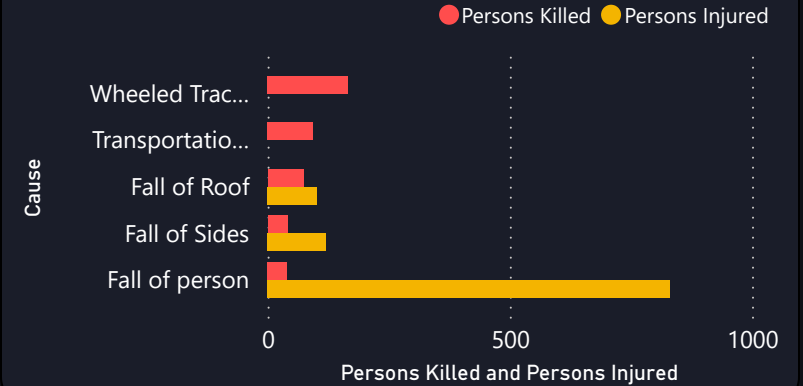
TOP 5 cause of FATAL Accidents (2013-2022)



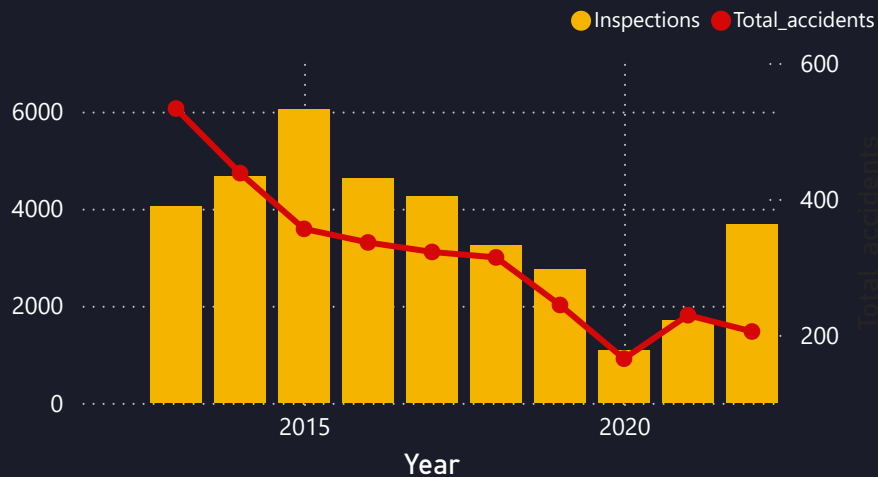
Top 5 cause of SERIOUS Accidents (2013-2022)



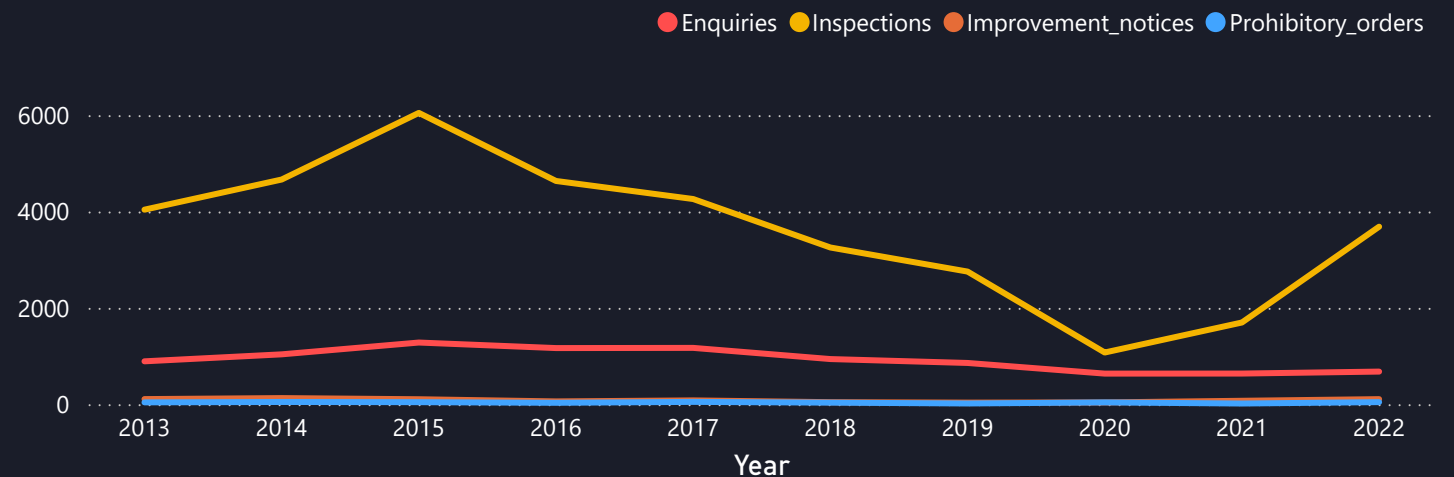
Total persons affected per cause (killed + injured)



DGMS Enforcement vs Total Accidents (2013-2022)



Enforcement Actions Trend (2013-2022)



Key Statistical Findings

Inspection vs Total Accidents
+0.64 (Moderate Positive)

Inspection vs Fatal Accidents
+0.36 (Weak Positive)

Prohibitory Orders vs Accidents
+0.36 (Weak Positive)





COMPANY BENCHMARK ANALYSIS (CIL SUBSIDIARIES) : 2016-2022

Comparative Assessment of Safety Performance Across Coal India Limited Subsidiaries

ECL (75)

Highest Deaths

NCL (21)

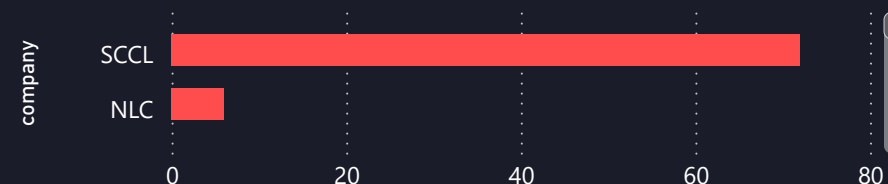
Lowest Deaths



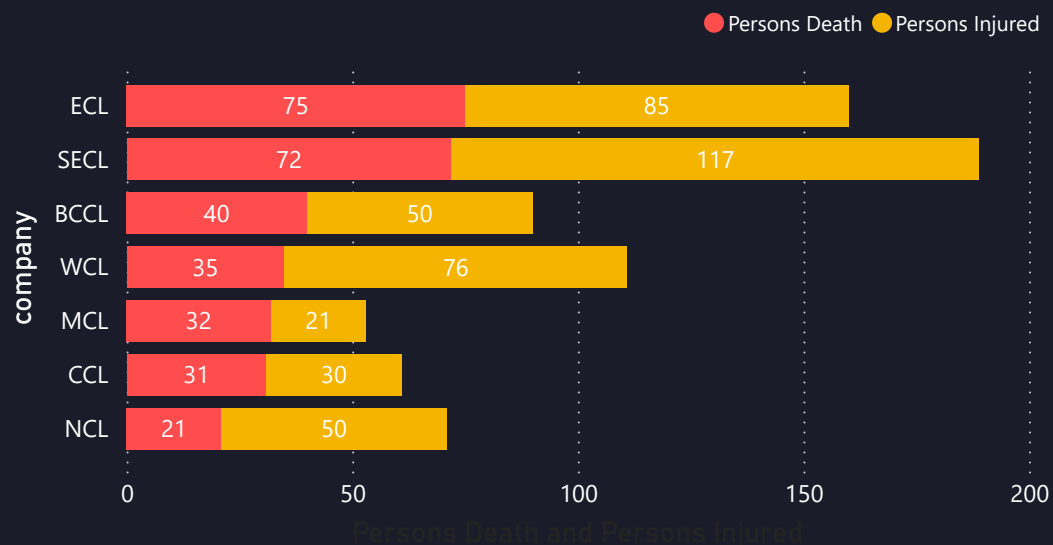
306

Total CIL Deaths

Total Deaths In Non-CIL Companies



Total persons affected by company (killed + seriously injured)



SECL (189)

Total Affected Persons

NON-CIL

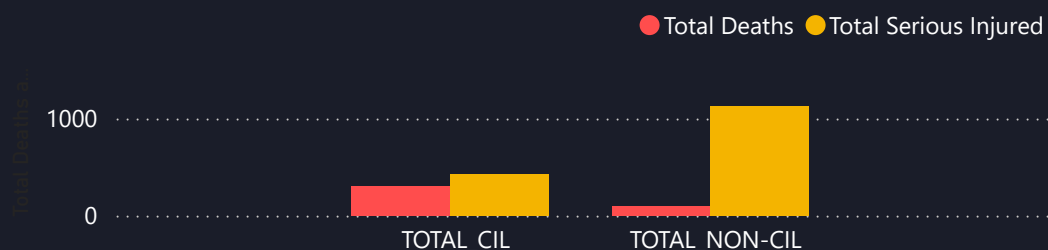
SCCL (1148)

Total Affected Person

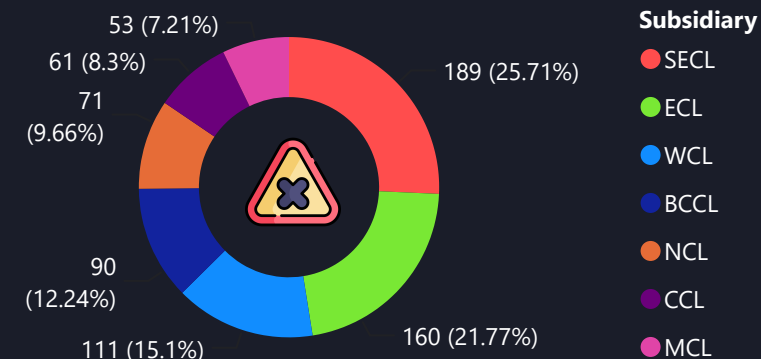
Yearly Death Trend per CIL Subsidiary

company	2016	2017	2018	2019	2020	2021	2022
WCL	6	6	6	4	5	6	2
SECL	11	10	16	8	12	7	8
NCL	4	4	4	3	2	3	1
MCL	2	5	8	10	5	2	0
ECL	35	10	2	8	8	10	2
CCL	6	8	8	2	2	2	3
BCCL	12	2	6	7	4	4	5

CIL vs Non-CIL Safety Benchmark (2016–2022)



Share of total harm by Subsidiary (2016-2022)





EXECUTIVE SUMMARY

Key insights derived from the analysis of Indian coal mine accident data (1997–2022).



NATIONAL SAFETY TRENDS

- Total accidents declined from 820 (1997) to 205 (2022)
- Fatalities reduced by 83.3% over the study period.
- Serious accidents peaked around 2005 and declined afterwards
- Underground mines accounted for over 50% of all fatalities.

CAUSE & ENFORCEMENT INSIGHTS

- Transportation-related incidents were the leading fatal cause.
- Roof falls remained a significant contributor to fatalities.
- Higher inspection activity showed moderate correlation with accident reduction.
- Enforcement actions were associated with improved safety outcomes.

COMPANY BENCHMARKING

- ECL recorded the highest fatalities among CIL subsidiaries
- NCL recorded the lowest fatalities
- SECL reported the highest number of persons affected
- Non-CIL (SCCL) companies experienced higher serious injuries than CIL companies

Source: **SANKET Statistical Analysis & Numbers: Key Evaluation of Trends in Coal Mine Accidents (2016-2022).**
Prepared using Python (Pandas), Power BI, and Exploratory Data Analysis.